

altairsim — A Universal I/O board on the Altair 680b

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The built-in `altair680` machine is a Motorola 6800 with just its onboard console — one 6850 ACIA at `F000/F001`. This example adds the **Universal I/O board** (`680uio`), the 680b's general-purpose expansion board: a **second 6850 serial port** and a **6820 PIA parallel port**, memory-mapped like everything on the 6800.

```
cd examples/altair680
altairsim altair680-uio.toml
```

You land at the `MON680` prompt, exactly as the plain `altair680` does — you are still typing at the **onboard** console. The UI/O is the *second* set of ports, sitting quietly until the guest talks to it.

Where the board lives

At its default S9 window (base `F000`) the board decodes:

Addresses	What
<code>F006</code> / <code>F007</code>	6850 ACIA — serial control/status and Rx/Tx data (<code>serial</code>)
<code>F008</code> – <code>F00B</code>	6820 PIA-C — sections A/B, control and data/DDR (<code>p1a</code> , <code>p1b</code>)
<code>F003</code>	switch inputs (<code>sense</code>) — fixed, read-only
<code>F010</code> – <code>F013</code>	8-bit non-latched output, Drive 1 + Drive 2

`SET uio0 base=0xF020` slides the serial+PIA window (the fixed `F003` and `F010` – `F013` do not move); `SET uio0 pias=2` populates a second 6820 (`p2a` / `p2b` at `F00C` – `F00F`). From the `altairsim>` monitor (press `^E` at the console), `SHOW MAP uio0` prints the live map.

Reaching the second serial from the monitor

The config wires the UI/O's serial line to `out:uio-serial.log` and PIA section `p1a` to `out:uio-pia.log` (both beside this file). `MON680's` `M addr` command examines and deposits a byte at an address, so you can drive the board by hand:

1. M F006 and deposit 03 (ACIA master reset), then M F006 again and deposit D1 ($\div 16$, 8N2) — the 6850 is not auto-configured on power-up.
2. M F007 and deposit a byte — it is transmitted, and appears in **uio-serial.log**.

The PIA works the same way: with control bit 2 set (deposit 04 to F008), a byte deposited to F009 is driven onto section A's lines and lands in **uio-pia.log**.

The two *.log files are created when the machine starts and are not part of the repository — they are just where this example's output goes.

See `reference/Altair 680b Universal IO Board.md` for the full register model, and the built-in `altair680` machine for the base machine this extends.